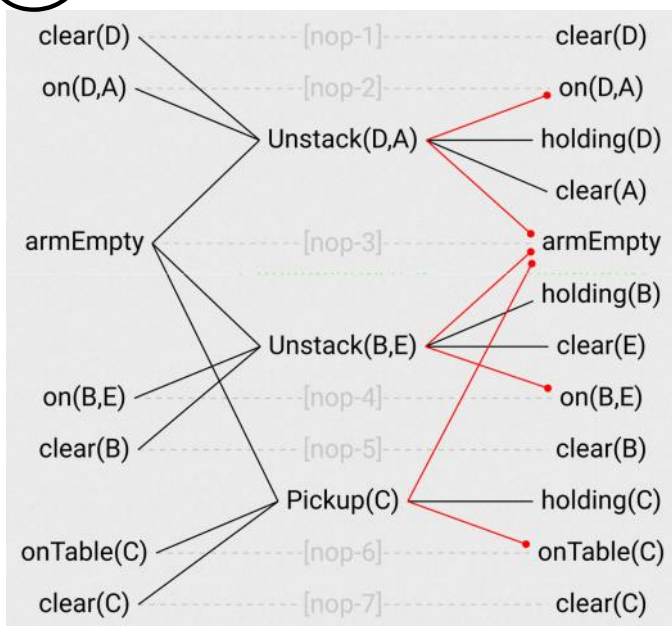


MUTEX CONDITIONS

- ① Competing Needs - For actions a & b whose preconditions are pairwise mutex.
- ② Inconsistent effects - If any precondition is a +ve effect of an action and a -ve effect of another parallel action.
- ③ Interference - If an action needs a pre-condition while the pre-cond. is being destroyed by another action.
- ④ $p \in \text{pre}(a) \wedge p \in \text{effects}^+(a)$ while $p \in \text{pre}(b) \wedge p \in \text{effects}^+(b)$
 a & b are mutex.

①

PA (Graphplan)



1) nop-3 , $\text{Unstack}(B,E)$ are mutex because -

- i) Inconsistent effects
(arm-empty)
- ii) Interference
(arm-empty)

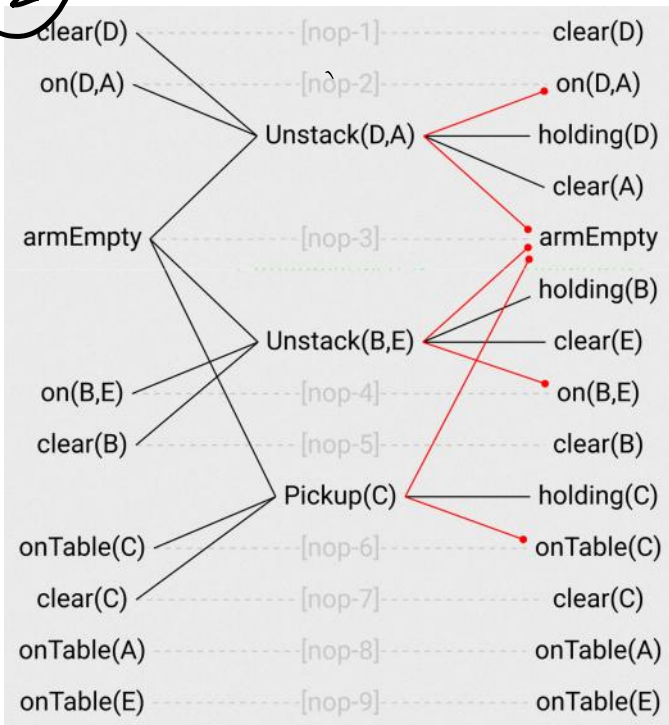
2) $\text{Un}(B,E)$, $\text{Pick}(C)$ are mutex



2) $Un(B,E), Pick(c)$ are mutex
because -

- i) $aE \in pre(Un) \ \& \ aE \in effects^-(Un)$
 $aE \in pre(Pi) \ \& \ aE \in effects^-(Pi)$

2



1) $cl(E), h(c)$

$\langle Un(B,E), Pi(c) \rangle$ **Mutex**

2) $cl(B), h(c)$

$\langle nop-5, Pi(c) \rangle$ **not mutex**

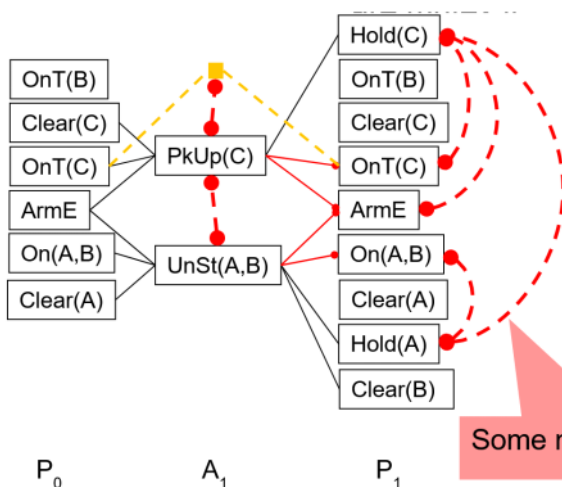
3) $cl(D), h(c)$

$\langle nop-1, Pi(c) \rangle$ **not mutex**

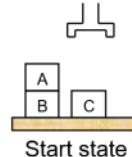
4) $h(B), h(c)$

$\langle Un(B,E), Pi(c) \rangle$ **Mutex**

Mutex relations



Two propositions $p \in P_i$ and $q \in P_j$ are mutex if **all** combinations of actions $\langle a, b \rangle$ in A_i with $p \in effects^+(a)$ and $q \in effects^+(b)$ are mutex.

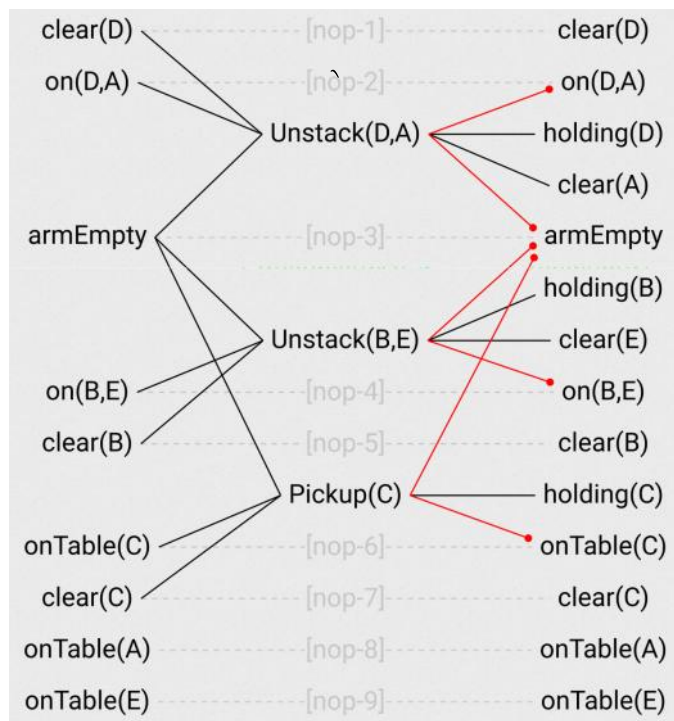


Some mutex propositions

3



① $pre(P_i(A))$



$$\textcircled{1} \text{pre}(P_i(A))$$

$$= \{aE, cl(A), onT(A)\}$$

pairs -

$$\textcircled{i} \langle aE, cl(A) \rangle - \text{Mutex}$$

$$\langle cl(A), onT(A) \rangle - \text{Not Mutex}$$

$$\langle onT(A), aE \rangle$$

$$(ii) \text{pre}(P_i(C)) = \{aE, cl(C), onT(C)\}$$

$$\langle aE, cl(C) \rangle - \text{Non Mutex}$$

$$\langle cl(C), onT(C) \rangle - \text{Non Mutex}$$

$$\langle onT(C), aE \rangle - \text{Non Mutex}$$

$$(iii) \text{pre}(P_d(C)) = \{h(C)\}$$

$$(iv) \text{pre}(Stack(B,E)) = \{h(B), c(E)\}$$

$$\langle h(B), c(E) \rangle$$

$$(v) \text{pre}(Stack(C,E)) = \{h(C), c(E)\}$$

$$\langle h(C), c(E) \rangle - \text{Mutex}$$

★ Two prepositions are mutex only if all pairs of all actions adding them to the layer are mutex.

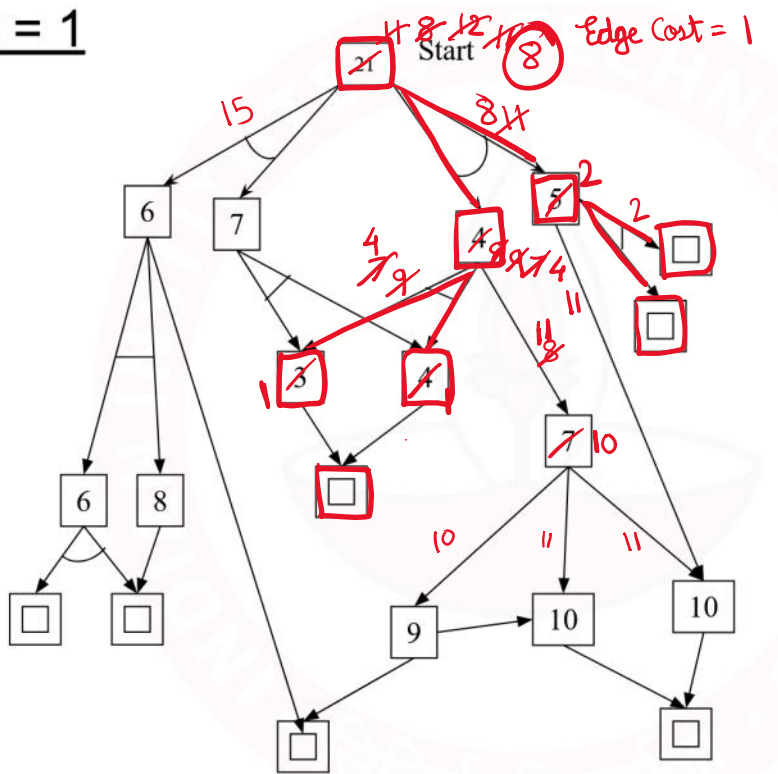
★ An action is applicable in Graphplan when -

- All pre-conditions are in the previous layer

★ An action is applicable to a system when

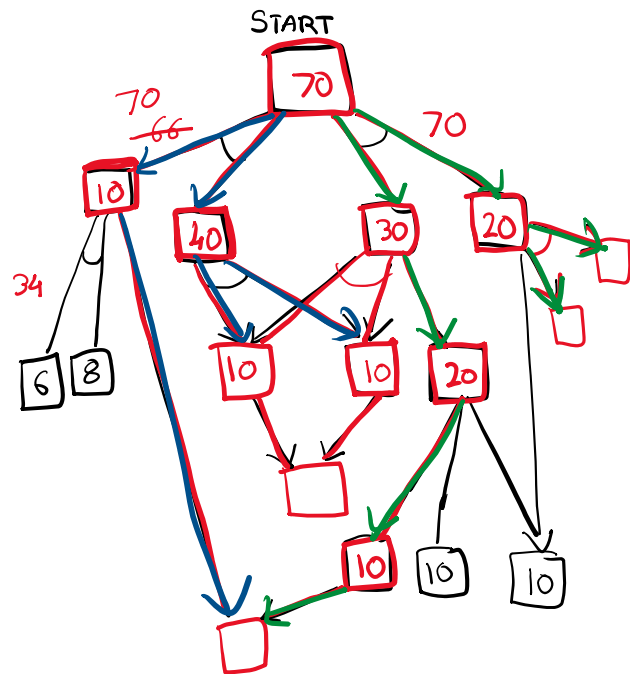
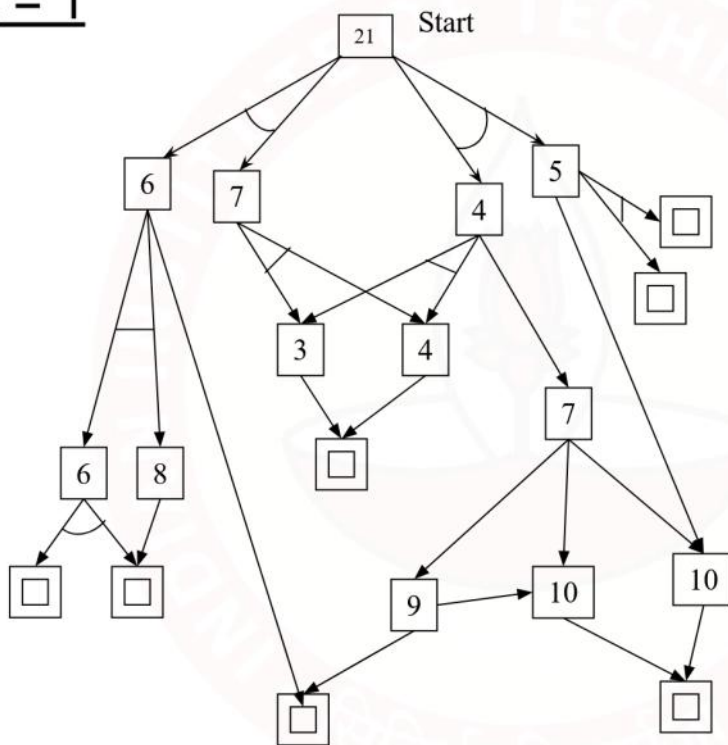
- All preconditions are in the previous layer
- No pair of preconditions is mutex
(All pairs of pre-conditions are non-mutex)

Cost = 1



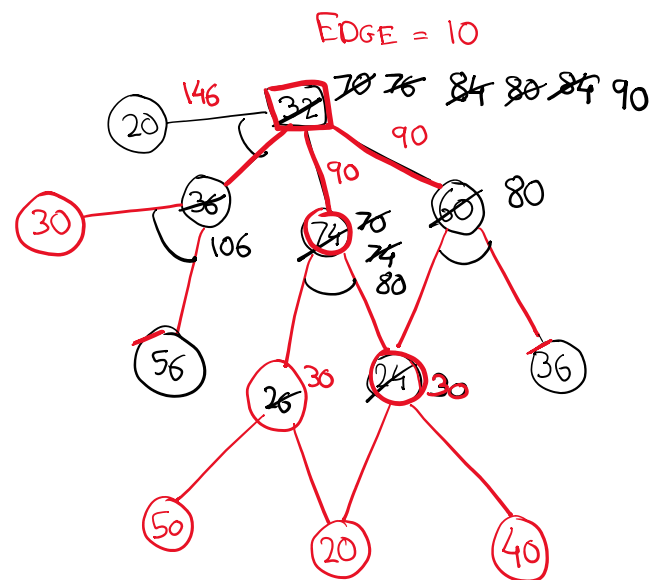
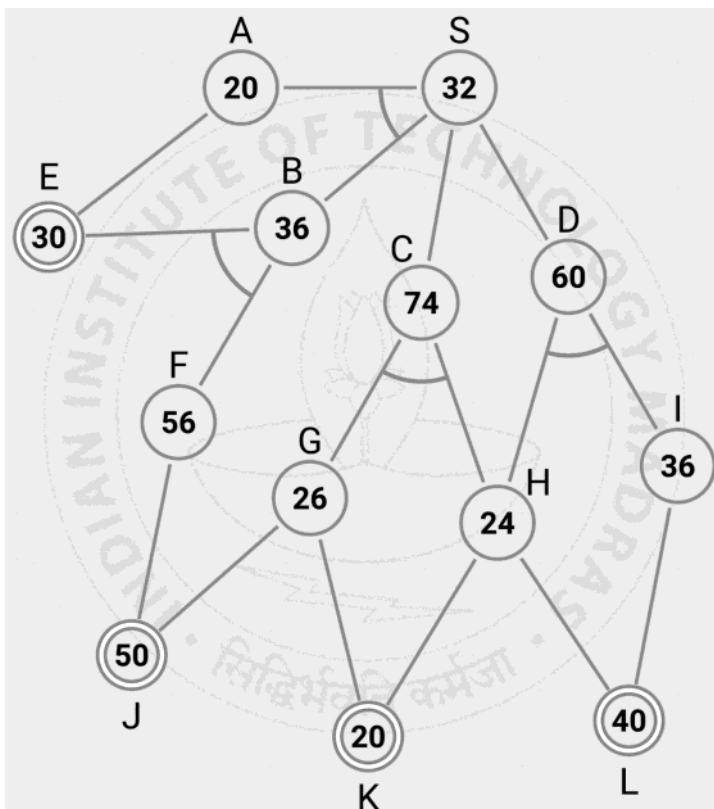
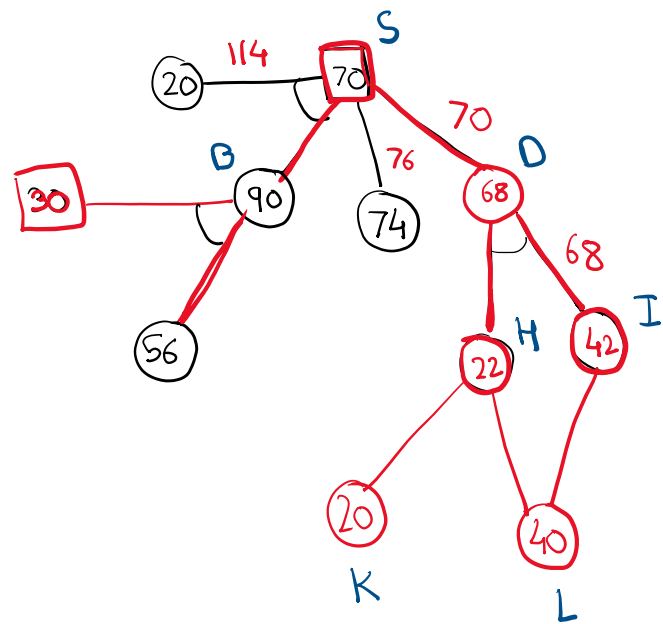
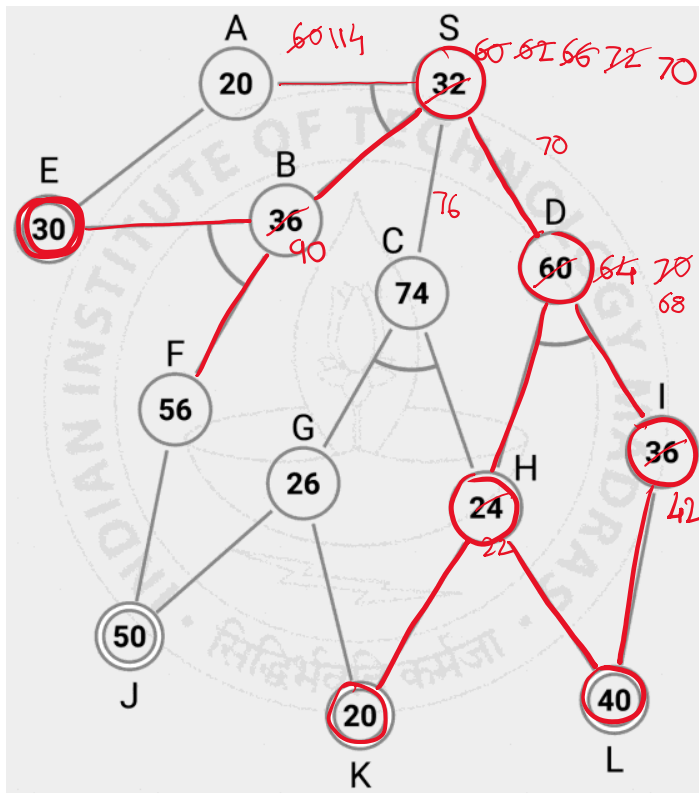
$G = \{\text{start}\}$

Cost = 1



PA - AO*

EDGE = 2



S, D, B, C, G, H